# Lab 7: ICMP & Firewalls

In this lab, we'll look at a specific type of ***denial-of-service*** attack known as ***ICMP flooding***, also known as the ***Ping of Death***. We'll look at some ICMP traffic on our own machines, and then we'll configure a ***firewall*** rule in ***Windows Firewall*** to disable this ICMP traffic.

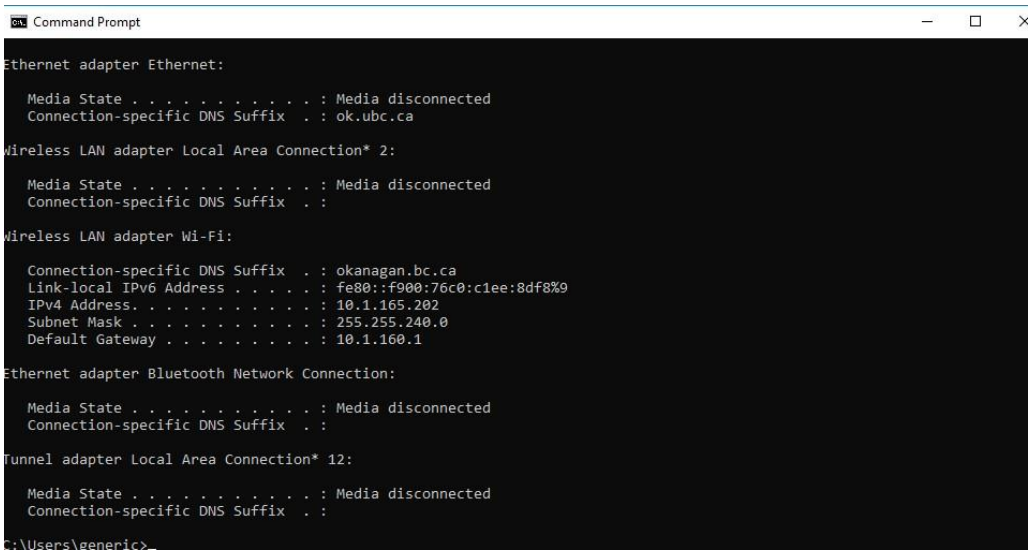
This lab assumes that you are working on a ***Windows*** machine. If you have one available, you should use that. Otherwise, please let me know if you are working on ***Mac*** or ***Linux***, as they have their own firewall utilities.

For this lab, ***you will submit a screenshot, as well as a text file (.txt) containing answers to questions in this lab***.

## Generating ICMP Traffic

First, we will need to generate some ***ICMP***, or ***Internet Control Message Protocol***, traffic. ICMP is typically used to ***ping*** another machine on a network to see if it is online, although ICMP can also be used for other diagnostic purposes – such as fetching timestamps.

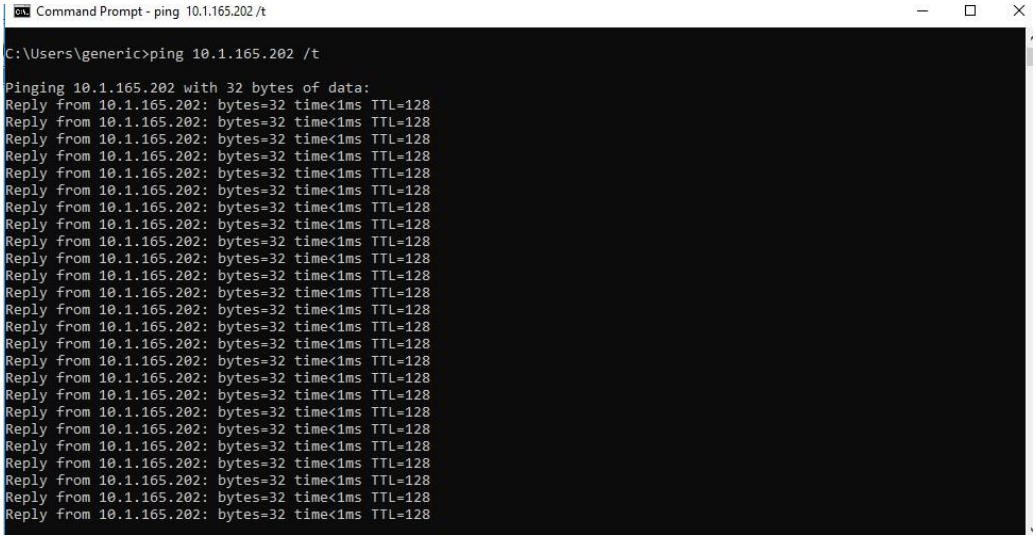
First, open a ***command prompt***, and use the ***ipconfig*** command like we did in the last lab to print out your network interface information. Find the connection that you are currently using to access the web, whether it is wired or wireless, and then find the ***IPv4*** address associated with it. This IPv4 address should be ***local***, meaning it starts with either ***10***, ***172.16***, or ***192.168***.



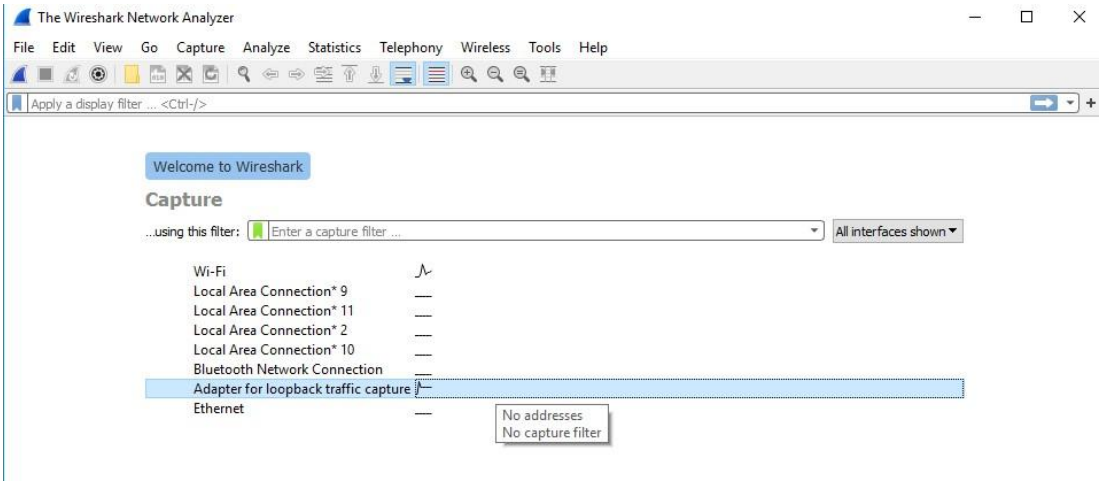

Once you have your local IP address, use the ***ping*** command in the command prompt to begin sending ICMP traffic to yourself. The syntax is ***ping <IP address> /t***, where the ***/t*** flag instructs ping to continue sending ICMP ping requests until it is ***interrupted*** (i.e. the command prompt is closed, or the user uses ***ctrl-C*** to kill the task).

This should result in a continual printout of ***ping replies***. They should originate from the same IP addresses and will probably have a very low ***time*** value.

***Question 1:*** The ***time*** value shows how long it took for the ***ping reply*** to be returned. In this case, your ***time values*** will be very small. Explain ***why*** this is. You may want to reference the ***network interface*** being used (mentioned down below).

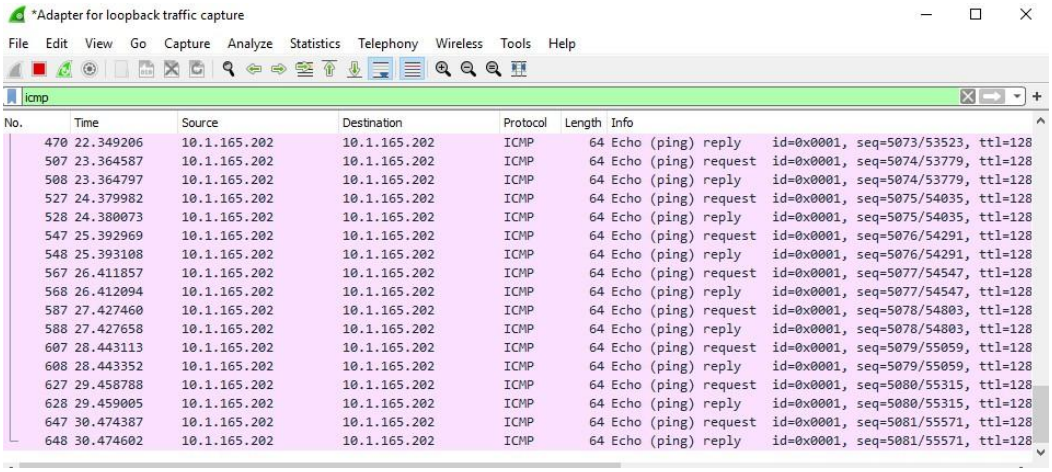



Now, let's go ahead and open up ***WireShark***. When it asks you to select a network interface, we will ***not*** be selecting your typical internet connection (WiFi or Ethernet). Instead, we'll be selecting the ***local loopback*** interface. The loopback interface allows traffic to be routed ***internally*** within your machine.




Once you've opened the loopback interface, begin ***capturing data***. You may want to filter by ***icmp*** to get rid of an extraneous requests or transmissions, so that we can look at ***just*** our ICMP traffic. You should see a long string of ***requests***, each with a subsequent ***reply***. Each of these represents a single ***ping*** exchange, with the time calculated based on how long it takes for the entire ***round trip*** of the request from sender to receiver, and then back again with the response.

Your traffic should look something like the image shown below:



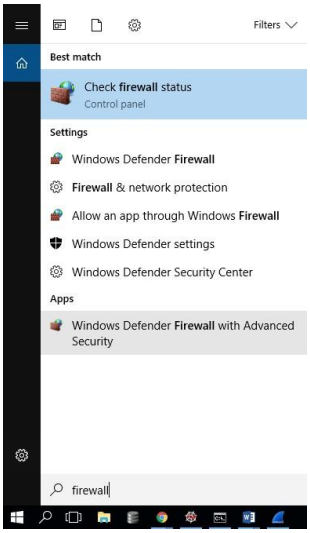

| No. | Time | Source | Destination | Protocol | Length | Info |
|---|---|---|---|---|---|---|
| 470 | 22.349206 | 10.1.165.202 | 10.1.165.202 | ICMP | 64 | Echo (ping) reply id=0x0001, seq=5073/53523, ttl=128 |
| 507 | 23.364587 | 10.1.165.202 | 10.1.165.202 | ICMP | 64 | Echo (ping) request id=0x0001, seq=5074/53779, ttl=128 |
| 508 | 23.364797 | 10.1.165.202 | 10.1.165.202 | ICMP | 64 | Echo (ping) reply id=0x0001, seq=5074/53779, ttl=128 |
| 527 | 24.379982 | 10.1.165.202 | 10.1.165.202 | ICMP | 64 | Echo (ping) request id=0x0001, seq=5075/54035, ttl=128 |
| 528 | 24.380073 | 10.1.165.202 | 10.1.165.202 | ICMP | 64 | Echo (ping) reply id=0x0001, seq=5075/54035, ttl=128 |
| 547 | 25.392969 | 10.1.165.202 | 10.1.165.202 | ICMP | 64 | Echo (ping) request id=0x0001, seq=5076/54291, ttl=128 |
| 548 | 25.393108 | 10.1.165.202 | 10.1.165.202 | ICMP | 64 | Echo (ping) reply id=0x0001, seq=5076/54291, ttl=128 |
| 567 | 26.411857 | 10.1.165.202 | 10.1.165.202 | ICMP | 64 | Echo (ping) request id=0x0001, seq=5077/54547, ttl=128 |
| 568 | 26.412094 | 10.1.165.202 | 10.1.165.202 | ICMP | 64 | Echo (ping) reply id=0x0001, seq=5077/54547, ttl=128 |
| 587 | 27.427460 | 10.1.165.202 | 10.1.165.202 | ICMP | 64 | Echo (ping) request id=0x0001, seq=5078/54803, ttl=128 |
| 588 | 27.427658 | 10.1.165.202 | 10.1.165.202 | ICMP | 64 | Echo (ping) reply id=0x0001, seq=5078/54803, ttl=128 |
| 607 | 28.443113 | 10.1.165.202 | 10.1.165.202 | ICMP | 64 | Echo (ping) request id=0x0001, seq=5079/55059, ttl=128 |
| 608 | 28.443352 | 10.1.165.202 | 10.1.165.202 | ICMP | 64 | Echo (ping) reply id=0x0001, seq=5079/55059, ttl=128 |
| 627 | 29.458788 | 10.1.165.202 | 10.1.165.202 | ICMP | 64 | Echo (ping) request id=0x0001, seq=5080/55315, ttl=128 |
| 628 | 29.459005 | 10.1.165.202 | 10.1.165.202 | ICMP | 64 | Echo (ping) reply id=0x0001, seq=5080/55315, ttl=128 |
| 647 | 30.474387 | 10.1.165.202 | 10.1.165.202 | ICMP | 64 | Echo (ping) request id=0x0001, seq=5081/55571, ttl=128 |
| 648 | 30.474602 | 10.1.165.202 | 10.1.165.202 | ICMP | 64 | Echo (ping) reply id=0x0001, seq=5081/55571, ttl=128 |

Look at the data for one of these ***requests*** or ***replies***, and use it to answer the following question:

***Question 2:*** What is the size of the ***frame*** (the entire message) for a typical ping request or reply? If an attacker wanted to produce network traffic equivalent to ***one gigabyte per second*** at a target, how many ***ICMP*** requests would they need to send per second (only counting requests, not responses)?

In a typical ICMP flood attack, you would see similar traffic, but in ***far greater numbers***, and likely originating from ***multiple source IP addresses*** (not just our own IP address). In fact, for most ***common denial of service attacks***, they will leave a blatant traffic signature that can be seen in ***WireShark***, typically as a massive number of incoming requests from one or more external IP addresses. If you believe you're being targeted by a ***denial-of-service*** attack, using WireShark is one of the easiest ways to confirm this.

***Question 3:*** Based on our discussion of ***DoS*** attacks in class, provide an example of ***at least one other*** type of network traffic you might capture in ***WireShark*** during such an attack. ***Specify what type of protocol it might be***, and what would the traffic ***look like*** (any abnormalities?).



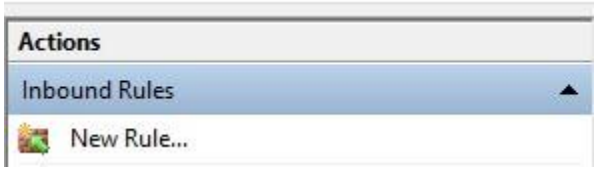

## Setting Up a Firewall

Now, let's go ahead and try setting up a ***firewall*** to help prevent an ***ICMP*** flood attack. Luckily, almost every operating system these days ships with a built-in, ***operating-system level firewall*** that can be used to allow or block traffic based on certain user-defined ***rules***.

Open a search bar and search "***firewall***" – you should see ***Windows Defender Firewall with Advanced Security*** pop up in your search results. This is the built-in firewall utility for Windows, and it will

allow you to define ***custom rules*** to handle ***inbound and outbound traffic***.

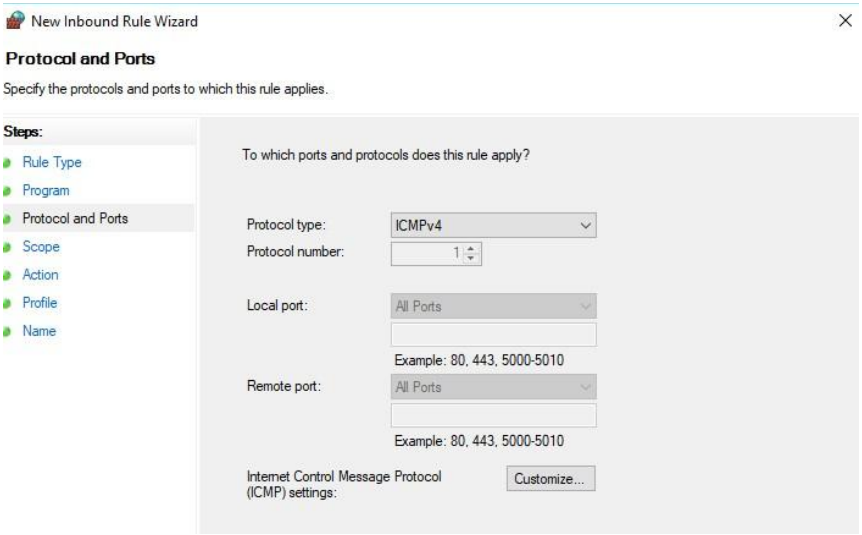
Go ahead and open the ***Defender Firewall with Advanced Security*** utility. It should present you with a new window, showing all ***inbound*** rules currently configured. Most of these are automatically generated when you install software to allow for updates, remote database connections, etc. Select the ***New Rule…*** menu item under ***Inbound Rules*** to begin defining a new inbound firewall rule:




This will take you through a short step-by-step ***wizard***. Make sure that you get the settings here correct:

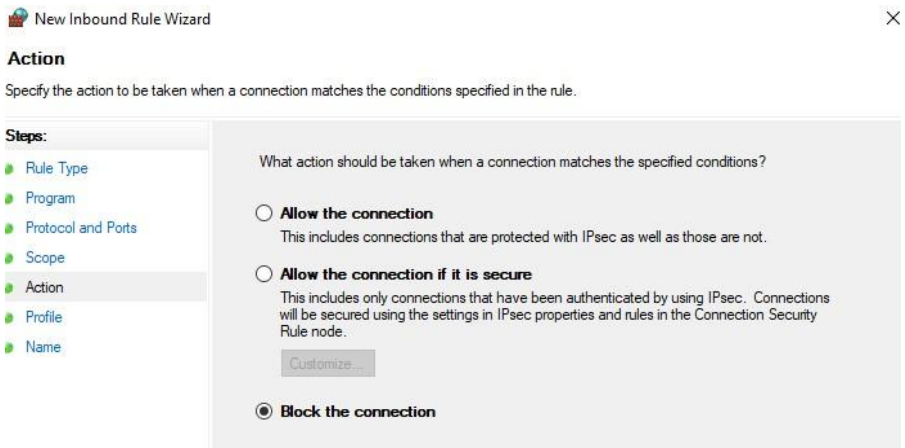
- It should be a ***Custom rule*** (not Program, Port, or Predefined)
- It should apply to ***all programs***

When you get to the ***Protocol and Ports*** step, you should select the ***ICMPv4*** protocol. ICMP is a ***portless*** protocol, so there will be no local or remote ports to enter at this step:



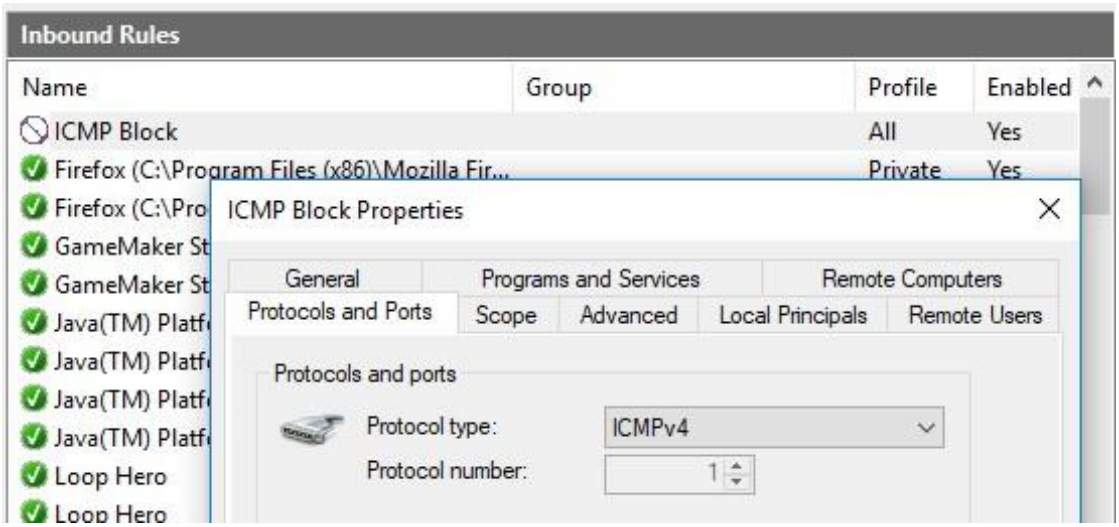

Then, you should simply continue with these settings:

- The scope should apply to ***any IP address***, both ***locally*** and ***remotely***
- The action should ***block the connection***
- The profile should ***apply this rule on all networks*** – this means it should be applied on all networks your computer is connected to – public, private, or work.




Finally, give your firewall rule an appropriate ***name*** and brief description. You should make sure this name is accurate and memorable, as there are ***many*** firewall rules and it is easy to "lose" your rule in the mix if you ever want to disable it.

***Congratulations***, you've created a firewall rule that should block all incoming ***ICMP*** traffic to your machine. ***Take a screenshot*** that looks like the following, showing both your ***rule in the list***, and the ***details pane for the rule (double click to open, it should be on the Protocols and Ports tab***):




Finally, answer these last two questions, both of which relate to the ***firewall***. You may want to refer to the ***slides from our previous lecture*** where we briefly discussed ***firewalls*** in the context of ***denial-of-service attacks***.

***Question 4:*** With our firewall rule, we have blocked ***all*** inbound ICMP requests on our machine. Is this a good idea? Explain ***why or why not***, and specifically link it back to the larger problem of ***blocking denial of service attacks*** – why is it ***difficult?***

***Question 5:*** Does our firewall rule completely protect us from an ***ICMP*** denial of service attack? ***Explain*** how an ICMP flood could still ***impact us as a user***, even with this firewall rule in place. (hint: where does the firewall ***end?***)

***Question 6:*** Try pinging yourself again and ***capturing the traffic in WireShark***. Does our firewall rule work? ***Provide*** a possible reason for this behaviour. It may be useful to check the ***Advanced > Interface Types*** section of the details pane for your firewall rule to get an understanding of ***why***.

## Lab Submission

***When you are done, submit your screenshot, along with your text file containing the answers to each of the six questions on Moodle***. You may also wish to ***disable*** (right-click "disable") your firewall rule when you are done, to prevent complications should you ever need to use ***ping*** again.